

# [Lecture Note] Quantitative Designs

GE2021 Research Technology, Spring 2025

Chad (Chungil Chae)

Fri, 21 March 2025

## Learning Objectives(Pajo, 2022)

- Describe the purpose of exploratory, descriptive, and explanatory studies.
- Compare and contrast cross-sectional and longitudinal studies.
- Explain the differences between nomothetic research and idiographic research.
- Discuss each type of experimental design and its advantages and disadvantages.

## SHORT VIDEO INTRODUCTION

<https://www.youtube.com/watch?v=VQbkw-SI8c4>

## Categorizations of Research Studies

### Major decisions for designing a study.

- Visualize and conceptualize the study.
- Determine the practical details of the study.
- Learn about the possible errors that could occur.
- Finally, begin working.

### The first characteristic of research design involves what we are trying to accomplish in this study.

- Exploring a new relationship between variables? Describing a phenomenon in detail? Explaining an established relationship?
- Common distinctions of studies are based on trying to explore, explain, or describe something.

**23 Quantitative or Qualitative**

- 24 • Quantitative.
  - 25 – Small samples; in-depth explorations; tends to construct a new theory.
  - 26 – Less design work, more data analysis.
  - 27 – Text, audio, image, and object analyses.
- 28 • Qualitative.
  - 29 – Large samples; clean measures to test hypothesis; tests an established theory.
  - 30 – Intensive design work, less data analysis.
  - 31 – Statistical analyses.

**32 Exploratory Studies—Answering “What?”**

- 33 • Exploratory studies are often used to investigate a new topic of research, a new thread of previously
- 34 established relationships, a new methodology, a new instrument of data collection, or to gain a deeper
- 35 understanding on a specific population.
- 36 • Useful as initiators for larger, complex studies. Aim at investigating, exploring, or figuring out a new,
- 37 innovative thread of knowledge.

**38 Descriptive Studies—Answering “How?”**

- 39 • Descriptive studies allow researchers to focus on describing a phenomenon or understanding the
- 40 details about people’s experiences of a particular event.
- 41 • Awareness of the difficulties that a specific group of people is experiencing may reveal possible ways
- 42 of intervening to help the group in need.
- 43 • Generally qualitative as it is often necessary to gather in-depth data and allow participants the flexi-
- 44 bility of expressing their feelings and situations in detail.

**45 Explanatory Studies—Answering “Why?”**

- 46 • Explanatory studies are reinforcing studies. They focus on explaining the reasons behind a phe-
- 47 nomenon, relationship, or event.
- 48 • Seek to explain the reasons behind an already known and/or established relationship.
- 49 • May be quantitative or qualitative depending on the researcher, but are usually quantitative.

## Cross-Sectional Versus Longitudinal Studies

### Cross-Sectional Studies

- Cross-sectional is the name for studies that collect data only once, a simple snapshot of data collected at one point in time.
  - For example, collecting information from participants who have purchased an expensive car in the last year.
- Refers to one shot of data collection, without any follow-up questions; useful when collecting data from a large number of participants.
- Difficult to establish any cause-effect relationships because all the variables are collected at the same time.

### Longitudinal Studies

- Longitudinal studies collect data at different points in time, which helps the data to be more accurate and avoid or minimize errors like inaccurate responses.
  - For example, we may want to know the long-term effects of a specific medication given to children.
- These different data collection points are called waves. The first data collection is often called base wave or wave I, typically continuing with wave II, wave III, and so on.
- Desirable for various topics as they are known for their accuracy.
- Types of longitudinal studies:
  - Panel studies follow and accumulate data from the same participants over a period of time.
  - Trend studies collect data at different points in time from different participants of the same population.
  - Cohort studies follows the same cohort of people over time. Includes a broader definition (such as being born, married, or giving birth in the same year) and are not the same people.

## Causality in Research

- An important feature of design is determining the type of relationship the study will investigate.
- Researchers often attempt to establish a causal relationship between variables of interest, which refers to attempts to determine that one or more variables (the independent variables), have caused the changes in another variable (the dependent variable).

## Nomothetic Research

- Nomothetic research aims at arriving at causal relationships that are easily applied to the population at large. Often quantitative because of its large scope.
  - Such as establishing that people with a higher education level will likely earn a higher salary.
- To establish nomothetic causality, a research study needs to establish three criteria:
  - Correlation: variables we are studying are related to each other; the changes in our independent variable should go hand in hand with changes in the dependent variable.
  - Time order: determines the actual order of changes from one variable to another; changes in the independent variable happen first in time and are accompanied by changes in the dependent variable happening later in time.
  - Non-spuriousness: A spurious relationship is explained away by a third variable that may have not been considered. Claiming nomothetic causality requires a non-spurious relationship.

## Idiographic Research

- Idiographic research attempts to explain the entire realm of influences and every possible detail that has influenced a particular phenomenon or event for an individual. Often qualitative in nature.
- It is time-consuming because it can investigate only one individual or event at a time because of its depth in truly capturing the information.
- Fuels later nomothetic research, but it is an important tool used to investigate a problem at an individual level before generalizing society at large.

## Experimental Designs

- An experimental design is a study that includes an experiment and then analyzes the results.

### Classic Experimental Design

- Experimental design refers to the presence of a test, experiment, intervention, or some form of manipulation happening at the time of data collection.
- Three requirements must be fulfilled in order to conduct an experimental design:
  - (1) the presence of a control group and an experimental group, (2) randomly assigning participants to the control and experimental groups, and (3) collecting data before and after the experiment, also known as pre- and posttesting.
- Experimental and Control Groups
  - The experimental group is undergoing some form of experimentation, such as training, taking a test or a drug, or some other form of intervention.

- The control group do everything the participants in the experimental group do, but are not given any test, drug, intervention, or manipulation, although they may be told that they are.
- The control group controls for confounding or disturbing variables, giving researchers confidence that findings are accurate and the outcome was a consequence of the experiment rather than extraneous reasons.

- Random Assignment

- When ready to collect data, first one should randomly place the participants into the experimental group and the control group.
- Minimizes the potential of characteristics to be the determining factor.
- Random select can be accomplished in a number of ways including drawing numbers or using a computer number generator.
- The researcher is usually aware of which group participants are assigned, whereas the participants are not aware. In instances where neither is aware, the studies are known as double-blind studies.

- Pre- and Posttesting

- Researchers need to pretest the participants and posttest them to figure out the actual results of our study, otherwise they may not notice any changes.
- The pretest and the posttest cannot be different or it may manipulate the outcome.

## Solomon Four-Group Experimental Design

- The Solomon four-group experimental design is a type of experimental design that helps to minimize measurement errors.
  - Protects the study from the biases of pretest.
  - Uses two experimental groups and two control groups.
  - One experimental and one control group take the pretest, the other two groups are subjected only to the intervention or the placebo and do not take the pretest.
  - The differences in results can clearly show whether the pretest is influencing the results of the study.

## Quasi-Experimental Designs

- Quasi-experimental designs conduct some form of intervention, testing, modification, or manipulation and examine the results.
- These studies are more relaxed in terms experimental design requirements such as randomly selecting participants for the control and the experimental groups or conducting a pretest and a posttest.
- Common quasi-experimental designs:
  - Randomized One-Group Posttest-Only.
  - Randomized Posttest-Only Control Group.
  - Non-Random Posttest-Only Control Group.

- Non-Random Pretest Posttest Control Group.
- Non-Random One-Group Pretest Posttest.

### Non-Experimental or Pre-Experimental Designs

- Non-experimental designs do not include any manipulation, such as training, intervention, or drug taking.
  - Crucial for researchers who are exploring something innovative and will then move into conducting experiments later.
  - Common designs include the Non-Random Cross-Sectional Survey and the Longitudinal Cohort Study.
- The term pre-experimental implies that these types of studies are conducted before the researcher establishes the need for an experiment.

### Class Activities

- In a small group, read the three scenarios presented below. Discuss whether the associations observed suggest that there is a causal relationship between these variables. Explain your answers.
  - Researcher A finds an association between the consumption of peanut butter and number of fatal accidents in the state of Massachusetts. He concludes that eating peanut butter causes fatal accidents.
  - Researcher B finds an association between ice cream eating and sunburns. She concludes that eating ice cream causes people to have more sunburns.
  - Researcher C is studying the effects of school lunches on student body mass index. School A notices a decrease in studying body mass index in January 2019. In this same month, School A made dramatic changes to their school lunch program, offering more healthy options. Researcher C concludes that changes in the school lunch program in January 2019 caused changes in student body mass index in January 2019.

- 
- You are interested in researching the effectiveness of a treatment drug for rosacea, a skin condition characterized by facial redness. Design an experimental study to test whether this treatment is effective for people who suffer from rosacea. More specifically, discuss:
    - What is your independent variable?
    - What is your dependent variable?
    - What is your null hypothesis?
    - What is your alternative hypothesis?
    - What are some ethical considerations related to conducting this study? How will you address any ethical considerations?

180

181

182

183

184

- 
- You are interested in studying changes in voting patterns among 18–24 year olds in Massachusetts in 2020. You propose to use a longitudinal design and collect data at two time points (Time 1 and Time 2). There are three types of longitudinal designs, panel, trend, and cohort. For each type of design identify the following key elements:

| Study Type | Population | Sample at Time 1 | Sample at Time 2 |
|------------|------------|------------------|------------------|
| Panel      |            |                  |                  |
| Trend      |            |                  |                  |
| Cohort     |            |                  |                  |

185

## Reference

186

Pajo, B. (2022). *Introduction to research methods: A hands-on approach*. Sage.